ISBI 2027 BENCHMARK · PROGRESS UPDATE · 15 OCTOBER 2026

Which feature network makes a good perceptual loss for LDCT?

From Week 2 to Week 3: CT dose network v2, the closeout checks, three seeds, statistics and the screen test

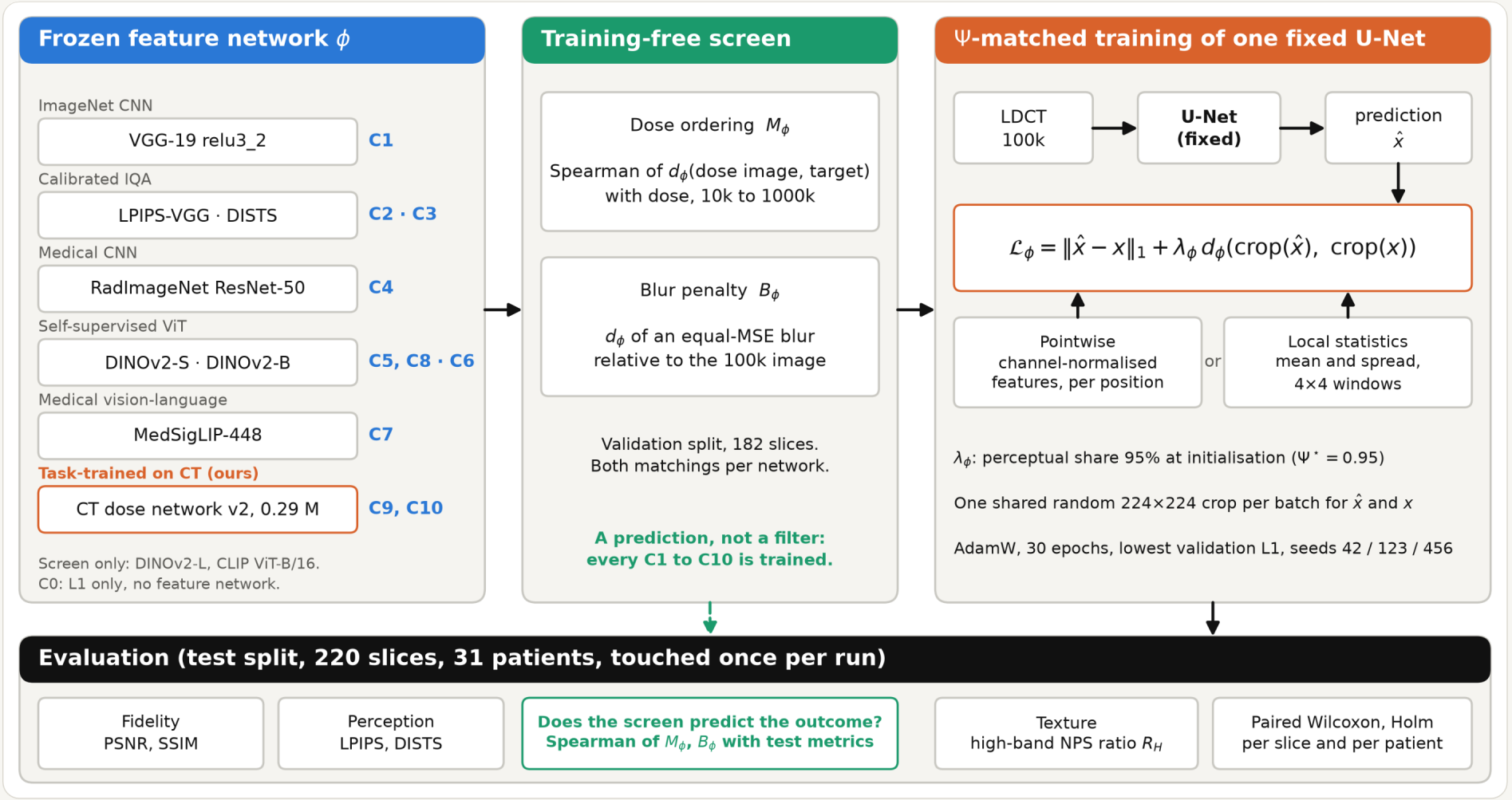
33 runs · 11 configurations × 3 seeds

Test split: 220 slices, 31 patients

Results freeze: 18 October

SETUP

One U-Net, only the loss changes



11
configurations

3
seeds

33
training runs

31
test patients

Fig 1 · C0 = L1 only · DINOv2-L and CLIP are screen only

SETUP · METRICS

What each test metric tells us

PSNR ↑

Pixel fidelity. Favours smooth images.

Input → C0 (L1)

23.04 → 30.92 dB

SSIM ↑

Local structure and contrast.

Input → C0 (L1)

0.742 → 0.869

LPIPS ↓

Perceived difference in AlexNet features.

Input → C0 (L1)

0.284 → 0.104

DISTS ↓

Structure and texture in VGG features.

Input → C0 (L1)

0.164 → 0.111

R_H → 1

High-frequency power, prediction ÷ NDCT. Below 1 = too smooth.

Input → C0 (L1)

6.17 → 0.66

Caveat

C2 trains on LPIPS-VGG, C3 on DISTS: partly scored by their own loss.

Marked * in Table 1

Ranking on grain uses the per-slice |1 – R_H| · reference values are three-seed means

v2: what changed and why

Item	v1	v2 (adopted)
Training target	6 dose classes	log10-dose regression
Val. accuracy	78.5%	78.7%
Screen M, pointwise	0.724	0.971
Screen M, statistics	0.712	0.852
Screen B, pw. / stat.	0.31 / 0.75	0.34 / 0.86

RESULT ON THE SCREEN

M rose by 0.25 and 0.14. B stayed below 1: blur still looks closer than noise.

Why

A graded target should make distance rise steadily with noise.

Unchanged

Same 3 conv blocks (287,328 parameters), same data.

Disclosure

Threshold lowered 0.9 → 0.8 after seeing the screen.

Validation screen, 182 slices · v2 best epoch 10 of 10: training probably cut short

v2: better screen, not better images

Test split, seed 42

Run	PSNR	DISTS ↓
C8	30.55	0.060
C9 v1 → v2	30.26 → 30.40	0.100 → 0.097
C10 v1 → v2	29.89 → 29.71	0.100 → 0.102

C9 a little better, C10 worse. Both far behind C8 (DINOv2-S, statistics).

Why: errors it under-weights

Error, equal MSE	vs noise
Blur	0.41 – 0.71
Brightness offset	0.16 – 0.35
Low-freq. error	0.16 – 0.26

Below 1 = cheaper than noise. Receptive field only about 23 px.

v2 frozen: matches grain statistics, blind to structure.

Diagnosis on CPU, 30 validation slices, pointwise matching, blocks 1 to 4

WEEK 2 · CLOSEOUT

Three checks before the new seeds

Variance fix

No more zero gradients in flat windows.

C10 v1: −0.28 dB, one seed only.

30.55 dB

C8 after the fix (30.53 before)

Reproducibility

C9 and C10 reproduce exactly from a clean commit.

Seed-42 C8 not replayed (disclosed).

0.0

relative difference to the log

C10 val. spikes

Cause: BatchNorm train/eval mismatch. Not a bug.

Saved epoch 17 is clean.

8 / 0 / 1

spike epochs, seeds 42 / 123 / 456

Seed 42 unless stated · decision: keep the protocol and disclose

Two more seeds, then the statistics

Runs

22 new runs at seeds 123 and 456. Frozen settings, all checks pass.

Data

220 test slices from 31 patients, averaged over 3 seeds.

Test families

A: each config vs C0. **B:** matching rule, model size.

Method	What it tells us
Paired Wilcoxon	Does A beat B on the same slices?
Holm correction	Guards against false positives from many tests
Rank-biserial r	How consistently one side wins (−1 to +1)
Patient level	Same tests on 31 patient means
Spearman ρ	Does the screen predict the trained ranking?

Families fixed before testing · about 3.2 A100 hours per seed

Table 1: three seeds, test split

Configuration	PSNR ↑	SSIM ↑	LPIPS ↓	DISTS ↓	R_H	|1 − R_H| ↓
LDCT input (100k)	23.04	0.7423	0.2843	0.1641	6.174	5.275
C0 (L1)	30.92 ± 0.07	0.869 ± .003	0.104 ± .001	0.111 ± .001	0.66 ± .03	0.35 ± .03
C1 (VGG-19, pointwise)	30.32 ± 0.22‡	0.861 ± .002	0.059 ± .001†	0.067 ± .002	0.80 ± .04	0.23 ± .03†
C2 (LPIPS-VGG, pointwise)	30.39 ± 0.31‡	0.863 ± .002	0.056 ± .005†*	0.065 ± .002	0.78 ± .01	0.24 ± .01†
C3 (DISTS, statistics)	29.81 ± 0.20‡	0.833 ± .024	0.056 ± .007†	0.066 ± .001*	0.82 ± .01	0.22 ± .02†
C4 (RadImageNet, pointw.)	28.70 ± 0.32‡	0.826 ± .008	0.087 ± .006	0.085 ± .001	0.80 ± .04	0.29 ± .02†
C5 (DINOv2-S, pointwise)	30.10 ± 0.51‡	0.863 ± .006	0.063 ± .008†	0.071 ± .007	0.68 ± .05	0.33 ± .04†
C6 (DINOv2-B, pointwise)	30.20 ± 0.44‡	0.862 ± .004	0.061 ± .003†	0.071 ± .003	0.73 ± .03	0.29 ± .03†
C7 (MedSigLIP, pointwise)	29.84 ± 0.39‡	0.860 ± .006	0.075 ± .011†	0.074 ± .007	0.53 ± .06	0.48 ± .06‡
C8 (DINOv2-S, statistics)	30.26 ± 0.28‡	0.861 ± .007	0.058 ± .011†	0.065 ± .005	0.86 ± .02	0.20 ± .01†
C9 (CT dose v2, pointwise)	30.44 ± 0.10‡	0.851 ± .022	0.067 ± .003†	0.097 ± .000	0.85 ± .02	0.20 ± .01†
C10 (CT dose v2, statistics)	29.80 ± 0.19‡	0.848 ± .004	0.072 ± .006†	0.098 ± .004	0.95 ± .06	0.17 ± .03†

± std over 3 per-seed means · † / ‡ better / worse than C0 (Family A, Holm) · * own training metric · bold = best

Comments on Table 1

What the table shows

- Every perceptual loss costs PSNR: 0.5 to 2.2 dB below C0.
- C1, C2 and C8 give the best trade-off: LPIPS 0.104 → about 0.058 for 0.5 to 0.7 dB.
- C10 and C8 have the lowest grain error, |1 – R_H|.
- Medical pretraining does not help: C4 has the worst PSNR, C7 is smoother than C0.
- The CT dose network gets the grain amount right, not the texture: DISTS 0.098 vs about 0.065.

What we cannot claim

- A DISTS winner: the top four are within one seed spread.
- A VGG advantage: LPIPS and DISTS are partly scored in VGG-like features (*).
- That a mean R_H near 1 is best: C10 scatters on both sides of 1.
- Small single-seed gaps: PSNR varies 0.2 to 0.5 dB over seeds with a perceptual term (0.07 for C0).

Exact values in the speaker notes

Which differences are real?

220 slices

Each slice counts as independent, but they come from 31 patients. p-values come out too small.

31 patients

Average each patient first, then test. Fewer points, but honest ones.

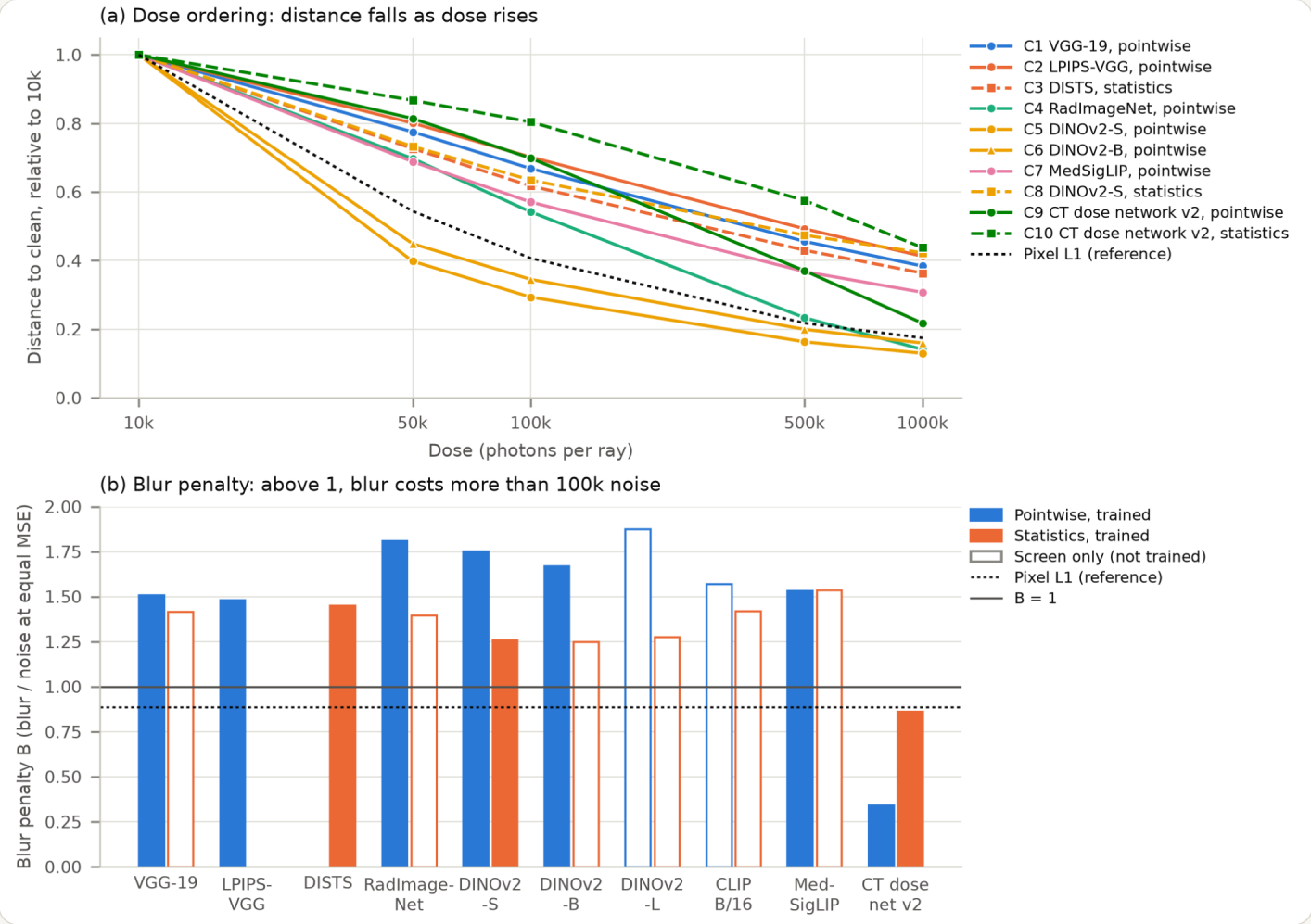
“Real”

Significant (Holm $p < 0.05$) at both levels, same sign at every seed.

Question	Compared	Slices	Patients	Answer
Does every perceptual loss differ from plain L1?	C1–C10 vs C0	29/30	30/30	Yes
Does statistics matching give DINOv2-S better grain?	C8 vs C5	yes	yes	Yes, all seeds
Does statistics matching cost the dose net PSNR?	C10 vs C9	yes	yes	Yes, 0.54 dB
Does the bigger DINOv2 give better grain?	C6 vs C5	yes	yes	Unclear: flips
DINOv2: do matching or model size change PSNR?	C8, C6 vs C5	yes	no	No
Dose net: does matching change LPIPS or grain?	C10 vs C9	yes	no	No
DINOv2: do matching or model size change LPIPS?	C8, C6 vs C5	no	no	No

Grain = $|1 - R_H|$ · matching = statistics instead of pointwise · model size = DINOv2-B instead of DINOv2-S

The training-free screen: M and B



M: dose ordering (a)

Does the distance fall as dose rises? 1 = always.

B: blur penalty (b)

Blur ÷ noise at equal MSE. Above 1 = blur is punished more.

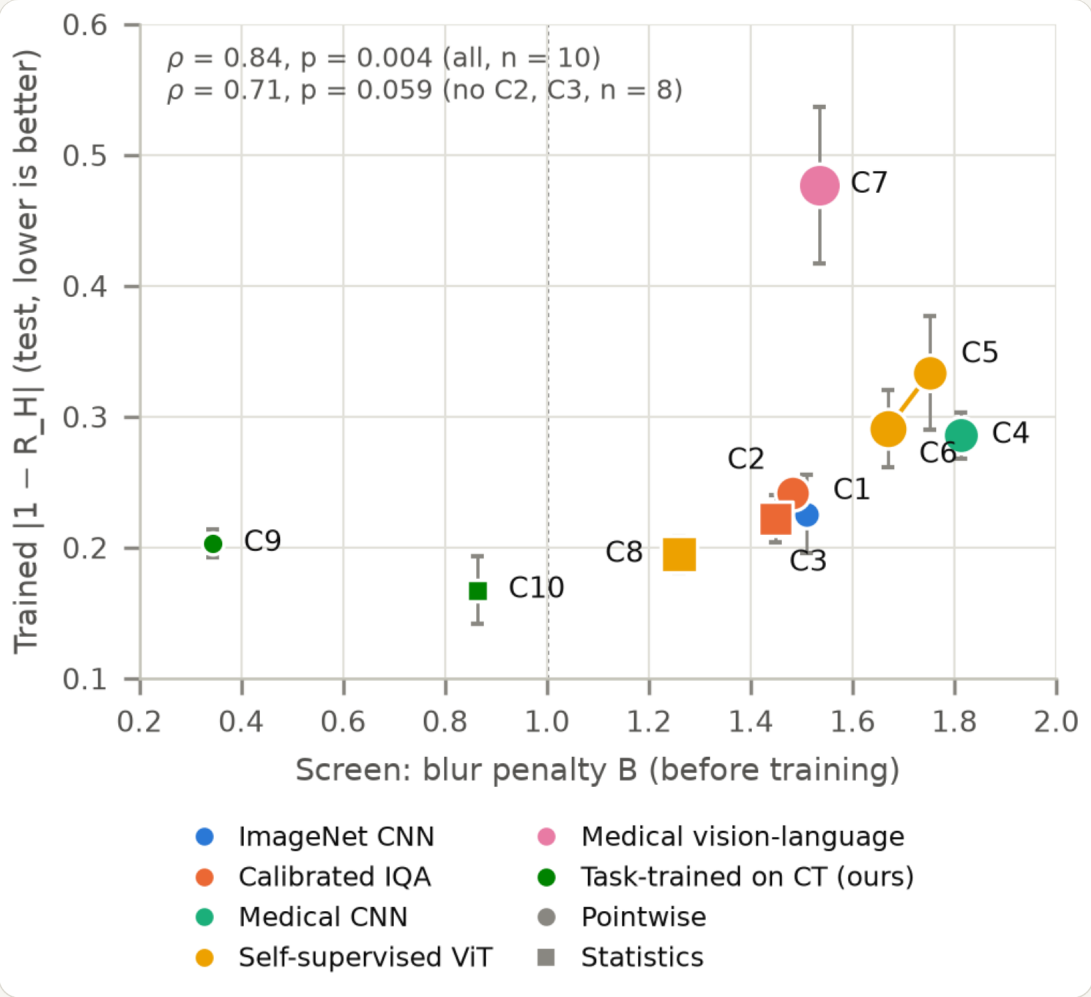
Reading

Pretrained: M ≈ 1 for all. CT dose v2: B below 1.

Fig 2 · validation split, 182 slices · DINOv2-L and CLIP are screen only

WEEK 3 · SCREEN

Does the screen predict the trained outcome? No.



Screen vs outcome	ρ	p
B vs |1 − R_H|	+0.84	0.0035
B vs DISTS	−0.06	0.89
M vs DISTS	−0.58	0.087

Wrong sign

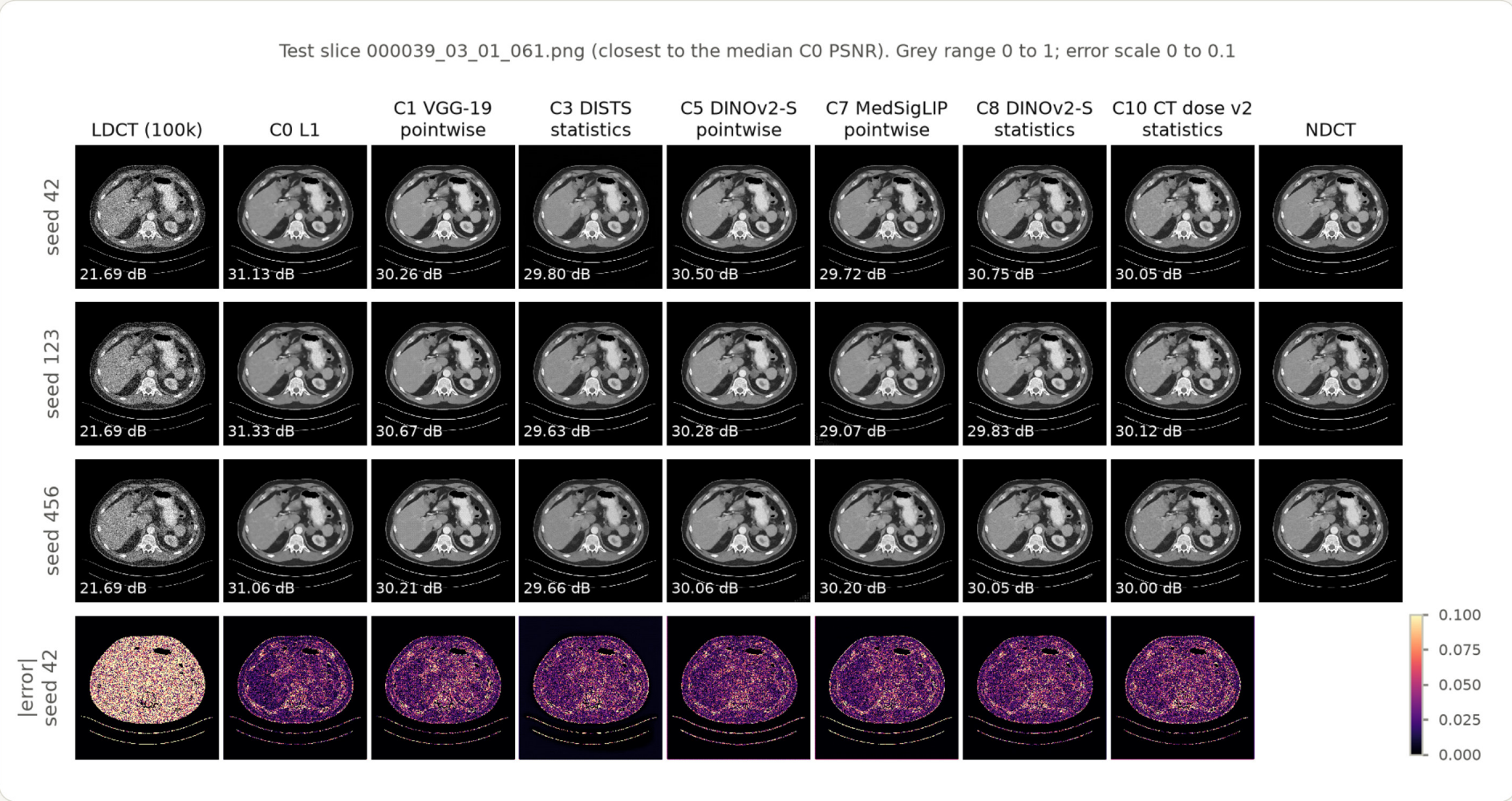
High B was meant to keep grain. Instead, high B goes with worse grain.

Likely confound

Statistics matching lowers B and improves grain at the same time.

Fig 4 · Spearman over 10 trained networks, permutation p, uncorrected

C0 is smoother; the perceptual losses differ subtly



Half the LPIPS for 0.5–0.7 dB

No clear winner on DISTS

Statistics matching helps grain

The screen does not predict

Fig 3 · slice 000039_03_01_061, chosen by rule · freeze 18 Oct · draft 23 Oct · deadline 26 Oct